Background-Input Leakage in Video Foreground Segmentation: An Empirical Audit and a Frozen Causal Protocol

Jichao Liu[1*]

[1*]Hainayun IoT Technology Co., Ltd., Qingdao, Shandong, 266103, China; ORCID: 0000-0001-5767-8654.

Corresponding author(s). E-mail(s): liujichao09@haier.com;

Abstract

Background-image guided networks are a popular paradigm for video foreground segmentation: a background image, often produced by a background-subtraction (BGS) algorithm such as SuBSENSE, is fed to a convolutional network together with the current frame, and is often trained with the same BGS masks as pseudo-labels. Its provenance—past frames, BGS-guided, or future-frame oracle—is rarely reported, making numbers incomparable. This paper is an empirical audit, not a new method. We formalize three leakage channels (background-input leakage, pseudo-label self-distillation, and future-frame oracle), define causal/leaky/oracle provenance levels, and introduce leakage-gap and oracle-gap metrics. Holding weights fixed and switching only the background source for a reproduced GNet on four held-out categories, the domain model drops from F1 = 0.6661 (leaky) to 0.4526 (causal), a relative drop of 32%, and remains below its SuBSENSE teacher (0.7256). The oracle gap is +0.159 for the domain model but only +0.021 for the synthetic model. Under a frozen causal protocol (stride 1), the synthetic background+CNN model (0.290 $\pm$ 0.041) does not outperform a non-learning causal residual threshold (0.392) or SuBSENSE (0.729); the same holds on SBI 2015 (0.347 vs. 0.589). A label-free causal teacher raises a same-domain model to 0.417, below its own teacher (0.465) and far below SuBSENSE; reliability-gating, boundary and degradation variants (0.240/0.249/0.260) show no consistent gain. Seed variance (0.041) exceeds most ablation differences. We recommend reporting provenance, teacher baselines, a frozen protocol and seed variance, and separating method gains from leakage. Code, video splits and per-video results are released.

Keywords: video foreground segmentation, background subtraction, background-input leakage, self-distillation, evaluation protocol, reproducibility

1 Introduction

Video foreground segmentation is a fundamental component of video surveillance, traffic monitoring and scene understanding. The CDnet 2014 benchmark [1] has made the task reproducible at the dataset level, and background subtraction (BGS) remains a strong and inexpensive baseline [2–5]. In recent years, a popular paradigm has emerged: a background image, usually produced by a BGS algorithm such as SuBSENSE [4], is fed to a convolutional network together with the current frame, and the network predicts a foreground mask. Representative methods include background-model-image networks [6], scene-specific and video-agnostic supervised models [7–13], and zero-shot background modeling [14]. The paradigm is intuitive: the background

provides a pixel-aligned reference, and the network learns where the current frame deviates from it.

The difficulty is that the reported accuracy of this paradigm conflates three sources of gain. First, the background input itself may contain the test-time BGS decision. If the background is updated only on pixels that SuBSENSE classifies as background, then the residual between the current frame and the background already approximates the SuBSENSE mask, so a network can reproduce the BGS decision without learning transferable foreground cues. Second, if the training labels are also generated by the same BGS, the network is distilled from its teacher; its domain-specific gain may then reflect self-distillation rather than a new mechanism. Third, if the background is estimated from the whole test sequence, for example as the temporal median of all frames or as the static background distributed with a dataset, the evaluation uses future frames and therefore constitutes an oracle. We refer to these three phenomena as background-input leakage, pseudo-label self-distillation, and future-frame oracle, respectively.

These channels are not exotic corner cases. They are common consequences of standard engineering choices: BGS-guided background updates are natural when a BGS mask is available, pseudo-labels are natural when manual annotation is expensive, and whole-video medians are natural when a static background is available. What is often missing is the provenance of the background in the reported result. Without provenance, a number such as "F1 = 0.96" cannot be compared with a number obtained under a causal protocol, because the two numbers may measure different quantities. This is the problem we address.

We do not propose a new segmentation network. Instead, we treat a reproduced GNet as a controlled testbed and ask what happens when only the background source changes. We define three provenance levels—causal, leaky and oracle—and two descriptive metrics: the leakage gap Δ_{leak} and the oracle gap Δ_{oracle}. The former quantifies how much performance is available through BGS-guided background updates; the latter quantifies how much performance is available through future frames. We then measure both gaps with fixed weights, fixed frames and fixed post-processing, so that the only variable is the background estimator. We complement this controlled comparison with a strict-causal evaluation against non-learning baselines, a multi-seed variance analysis, a cross-dataset check on SBI 2015, and a released reproducibility package.

Our contributions are audit-oriented:

1. We formalize three leakage channels in background-guided video foreground segmentation and define causal/leaky/oracle provenance levels together with leakage-gap and oracle-gap metrics (Section 3).
2. We perform a same-weight, provenance-only switching experiment and quantify the dependence of a domain-trained model on BGS-guided backgrounds (-0.214 F1) and on future-frame backgrounds ($+0.160$) (Section 4.2).
3. We report a strict-causal negative result: a synthetic background+CNN pipeline (0.290 ± 0.041) does not outperform a non-learning causal residual threshold (0.392) or SuBSENSE (0.729), and the same qualitative conclusion holds on SBI 2015 (Section 4.4, Section 4.6). A label-free causal teacher yields 0.417 on the same four held-out categories, below the SuBSENSE teacher (0.726), and reliability, boundary and degradation variants do not improve on the synthetic baseline (Section 4.4, Section 4.7).
4. We show that seed variance (0.041) exceeds most ablation differences and that a reliability-gating variant does not consistently improve accuracy, and we distill these findings into a reporting checklist, and we release the code, splits and per-video results (Section 4.5, Section 4.7, Section 5).

We are explicit about what we do not claim. We do not claim state-of-the-art accuracy, we do not claim that our testbed covers all architectures or background estimators, and we do not claim that every background-guided method leaks to the same degree. Our claims concern the provenance levels we can control and the models we can reproduce. The broader message is methodological: because provenance can change F1 by more than twenty points, it must be reported.

2 Related Work

2.1 Background-guided foreground segmentation

Background-model-image networks concatenate or fuse a background image with the current frame. Representative examples include Kim and Ha [6], scene-specific CNNs [7, 8], BSUV-Net [11] and BSUV-Net 2.0 [12], and BgSubNet [13]. Table 1 summarizes the provenance of representative methods: it ranges from BGS-generated and manually selected backgrounds to offline whole-video reconstruction and synthetic/paired backgrounds; two further methods (Braham [7]; Babaee [8]) do not state the source in the accessible materials. Provenance is therefore often implicit, and in several cases it is not stated at all. BgSubNet uses synthetic pretraining and semi-supervised fine-tuning, but the real-scene background source is not stated in the accessible abstract. By contrast, FgSegNet and FgSegNet v2 [9, 10], ZBS [14], motion-based video object segmentation [15], and BGS-only methods [16, 17] do not take a background image. Our claims therefore concern the background-image subset, not all video foreground segmentation; the design question is where the background comes from, and whether that source uses BGS decisions or future frames. The released audit file extends Table 1 to 14 methods, including frame-only, zero-shot, motion-based and BGS-only methods.

2.2 Unsupervised and self-supervised BGS

Unsupervised BGS ranges from Gaussian mixture and non-parametric models [2, 19, 20] to sample-based and self-organizing methods [3–5, 21]; see [22–24] for surveys. Recent examples include generator-discriminator learning [17] and tensor RPCA [16]. These methods are usually evaluated as standalone BGS. When their output becomes a network input, the provenance question reappears: the network may exploit the estimator's decision, and the gain may not transfer. Our switching experiment exposes this dependence.

2.3 Input cues, shortcuts and over-reliance

The idea that a network can over-rely on a cue that is correlated with the label is well established. Shortcut learning [25] and the Clever Hans phenomenon [26] show that classifiers can exploit incidental correlations that do not transfer. In data mining and machine learning more broadly, the same failure mode is studied as leakage [27, 28]. Motion-as-Option [15] shows that optical flow is strongly correlated with foreground masks and that networks can over-depend on motion, becoming unstable under confusing motion. We study the same phenomenon for a different cue: the background input. In the leaky setting, the background is correlated with the BGS mask, and a network can exploit this correlation. The leakage gap is an operational measure of how much of the reported performance is consistent with such exploitation. Pseudo-label self-distillation is the related channel: a student can fit its teacher's errors [29, 30], so the teacher baseline is needed.

2.4 Background noise and uncertainty

Several works estimate the reliability of a reconstructed background. An autoencoder-based method predicts background noise for thresholding [18], and uncertainty estimation for deep networks is surveyed in [31]. These works motivate per-pixel reliability, but reliability is not the focus of our audit. We include a reliability-gated variant as a diagnostic and report a negative result: the variant does not improve accuracy under our protocol. This negative result is useful because it separates "the background is unreliable" from "a learned reliability gate improves segmentation"; the two claims are often conflated.

2.5 Evaluation protocols and reproducibility

CDnet 2014 [1], its earlier version [32], and SBI 2015 [33, 34] are widely used, but protocol details such as frame stride, shadow handling, region-of-interest masking and aggregation can change reported numbers; see [24] for an evaluation study. Segmentation measures themselves are not interchangeable, as discussed for boundary and region

Table 1 Background provenance of representative background-image methods. n.s.: not stated; BGS: background generated/updated by a BGS algorithm; manual: empty background selected from the test video; offline: uses the complete video; Code: public code/weights found (n.s. = not found/verified).

Method	Input	Background source	Online	Code
Kim [6]	BG + frames	SuBSENSE-gen.	BGS	n.s.
BSUV-Net [11]	frame + 2 BG + FPM	empty: selected median; recent: 100-frame median	manual + causal	yes
BSUV-Net 2.0 [12]	frame + 2 BG + FPM	manual empty + 100-frame median	manual + causal	yes
BgSubNet [13]	image + BG	BG image (source n.s.)	n.s.	n.s.
Autoencoder [18]	frame + recon. BG	complete-video AE + median	offline	n.s.

metrics in [35]. Reproducibility is further complicated when code, weights, splits or background estimators are unavailable [28], and seed variance is routinely under-reported [36–38]. Our protocol fixes these choices and reports them explicitly.

2.6 Video object segmentation benchmarks

Semi-supervised video object segmentation methods such as OSVOS [39] and the DAVIS benchmark [40] also produce per-frame foreground masks, but they assume a first-frame annotation and are evaluated with region and boundary measures designed for that setting. Our task is unsupervised background subtraction without a first-frame mask, so these methods are not direct baselines; we cite them only to delimit the scope of the audit.

3 Audit Design and Protocol

3.1 Background provenance levels

Let $I_{1:T}$ be a video and let B_t be the background used at time t. We distinguish three provenance levels; Figure 1 summarizes the controlled switching design and the channels attached to each level.

- **Causal.** $B_t = f(I_{1:t})$ with f using only current and past frames, and using neither segmentation labels nor BGS masks. A running average or a causal temporal median are examples.
- **Leaky.** $B_t = f(I_{1:t}, M^{\mathrm{BGS}}_{1:t})$, where M^{BGS} is a BGS mask. A typical example updates the background only on pixels classified as background. The background is still causal in time, but it carries the BGS decision.
- **Oracle.** $B_t = f(I_{1:T})$ uses future frames. The temporal median of the whole video and the static background distributed with a dataset are examples.

The term "leaky" is operational. Using a BGS output is not by itself an error: if the goal is to post-process a BGS result, a BGS-guided background is a legitimate design. The problem arises when a method claims to learn transferable foreground cues or cross-scene generalization while its background input carries the test-time BGS decision. In that case the reported gain and the claimed mechanism refer to different things.

3.2 Leakage channels

We distinguish three channels.

L1: background-input leakage. The background input encodes the BGS decision. A network can reproduce the decision from the input without learning foreground appearance.
L2: pseudo-label self-distillation. The training labels and the inference background come from the same BGS. The model is then distilled from its teacher, and its gain should be compared with the teacher's accuracy.
L3: future-frame oracle. The background uses frames after t. The evaluation measures a quantity that is unavailable online.

The channels can co-occur. A domain-trained model that is trained with SuBSENSE pseudo-labels and evaluated with a SuBSENSE-guided running background is exposed to L1 and L2 simultaneously. An evaluation that uses a whole-video median is exposed to L3. Our experiments are designed to separate L1/L3 from L2 where possible: in the provenance-switching experiment the weights are fixed, so any change is due to

the input channel; in the self-distillation test we compare the model with its teacher.

3.3 Gap metrics

For a fixed model and test set, we define

$$\begin{aligned} \Delta_{\text{leak}} &= \text{F1}_{\text{leaky}} - \text{F1}_{\text{causal}}, \\ \Delta_{\text{oracle}} &= \text{F1}_{\text{oracle}} - \text{F1}_{\text{causal}}. \end{aligned} \tag{1}$$

A positive Δ_{leak} indicates that performance increases when the background input carries BGS decisions; a positive Δ_{oracle} indicates that performance increases when future frames are available. Smaller gaps are desirable for a method that claims online, leakage-free operation. We emphasize that the gaps are descriptive: they quantify the exploitable dependence under the tested background estimators, not a causal effect of leakage in general. We report F1, percentage of wrong classifications (PWC) and boundary F1 (BF1), with F1 as the primary metric; BF1 follows the boundary-band evaluation discussed in [35]. For the boundary metric we use a 2-pixel tolerance band around the ground-truth contour.

3.4 Frozen causal protocol

All strict-causal results use the following protocol: all frames inside the temporal region of interest, stride 1; shadows counted as background; pixels outside the region of interest ignored; per-video F1 first, then the arithmetic mean over categories; decision threshold 0.5. The same weights, frames and post-processing are used across provenance levels; only the background estimator changes. Non-learning residual baselines scan a threshold τ on the test set and report the best category-average F1. This makes the non-learning baselines optimistic, which is deliberate: it prevents us from overstating the difficulty of the baseline. The controlled provenance-switching experiment in Section 4.2 uses the same stride-1 protocol on four held-out categories. All provenance levels in that experiment use the same weights and frames, so the only variable is the background source.

3.5 Controlled testbed

We reproduce a GNet-style network as a testbed: a ResNet101 dual encoder, an ASPP module, and a temporal self-attention module (TSAM) with

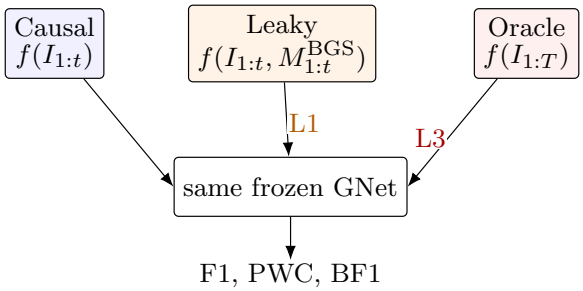


Fig. 1 Background provenance and the three leakage channels. The audit holds the network weights fixed and changes only the background estimator: leaky backgrounds carry the BGS decision (L1), training labels may come from the same BGS (L2), and oracle backgrounds use future frames (L3).

12 layers and 8 heads, operating on 513×513 inputs. The network takes the current frame and a background image and predicts a foreground mask at the same resolution. The *synthetic* model is trained on composited Cityscapes [41] and COCO [42] images; its background is a causal exponential moving average. The *domain* model is trained on seven CDnet categories and tested on four held-out categories; it follows the original paradigm, using SuBSENSE pseudo-labels and a SuBSENSE-guided background. The *reliability* variant adds a learned per-pixel gate on the feature difference; it is a diagnostic, not the main object of study. We also consider a *full* variant that adds boundary attention and background-degradation simulation, and a *label-free* variant trained with a causal-median teacher. These variants are diagnostics: as shown in Section 4.4, none of them improves on the synthetic baseline under the strict protocol. The testbed is audited, not proposed as a new method.

4 Experiments

4.1 Datasets, splits and implementation

The main evaluation uses CDnet 2014 [1]: 53 videos in 11 categories. The synthetic model is trained on Cityscapes and COCO composites and never sees a CDnet video. The domain model uses a category-held-out split: it trains on seven categories and tests on badWeather, nightVideos, turbulence and thermal, with no video overlap. The cross-dataset evaluation uses SBI 2015 [33, 34]; no

model is trained on SBI. Video-level split files and evaluation commands are released.

The network is trained with SGD at learning rate 0.01, momentum 0.9 and weight decay 5×10^{-4}, with cosine annealing, for 8 epochs of 600 steps, batch size 8. The seeds are $\{0, 1, 2\}$ for the synthetic model. The input size is 513×513. All reported numbers are category averages on the corresponding test sets.

4.2 RQ1: Does background provenance change results at fixed weights?

Table 2 reports the same weights evaluated with four background sources on the four held-out categories. The four sources are *leaky* (a causal running average updated only on SuBSENSE background pixels), *causal* (a mask-free causal running average), *first frame* (a static first frame), and *oracle* (the static background distributed with CDnet, which is essentially the whole-video temporal median; its mean absolute difference from the whole-video median is 0.22/255).

Four observations follow. First, the domain model loses 0.214 F1 (a relative drop of 32%) when the BGS-guided background is replaced by a mask-free causal background. The synthetic model loses only 0.031. The difference between the two models is consistent with the hypothesis that training with BGS-guided backgrounds teaches the network to exploit the BGS decision channel. The drops are sequence-dependent: thermal/corridor falls from 0.772 (leaky) to 0.181 (causal) and badWeather/blizzard from 0.654 to 0.286, while the camera-stable turbulence sequences lose about 0.10. Second, for the domain model the leaky background is even better than the oracle background (0.6661 vs. 0.6119). This is important: the oracle is not automatically the best input, because the model was trained on the leaky distribution. The result shows that training–test provenance alignment can create a gain that is neither oracle-based nor mechanism-based. Third, the synthetic model is almost much less sensitive to provenance (0.330–0.361), with $\Delta_{\text{oracle}} = 0.021$. This indicates that its gain does not come from the BGS channel, but it does not imply accuracy; its absolute performance is low, as shown next. A paired bootstrap over the 19 videos (20,000 resamples) gives a 95% confidence interval of $[+0.14, +0.28]$ for the domain model's leakage gap and $[+0.03, +0.27]$ for its oracle gap, both excluding zero. For the synthetic model at stride 1 the intervals are $[-0.01, +0.07]$ and $[-0.02, +0.05]$, consistent with no provenance effect.

4.3 RQ2: Is the domain gain distinguishable from self-distillation?

The SuBSENSE teacher reaches F1 = 0.7256 on the four held-out categories. The domain model reaches 0.6661 with the leaky background, which is below its teacher, and 0.4526 with the causal background. Thus the simple test "does the student beat its teacher after removing the BGS-guided input?" fails. The domain gain is therefore not evidence of a better mechanism; at minimum it is entangled with self-distillation and background-input leakage. This does not prove that no mechanism gain exists, but it shifts the burden of proof: a method that trains with BGS pseudo-labels should report the teacher baseline and the causal background result.

4.4 RQ3: Strict-causal baselines and the negative result

Table 3 reports the strict-causal comparison on all 53 CDnet videos at stride 1. SuBSENSE reaches 0.7287, close to the value of 0.7408 reported in a recent benchmark summary [14], which validates the evaluation implementation. The current-frame-only network reaches 0.2209, confirming that temporal background information is useful in principle. The synthetic GNet reaches 0.290 ± 0.041 over three seeds. A non-learning causal residual threshold reaches 0.3918, and a first-frame residual threshold reaches 0.3973. Both are higher than the synthetic network. An oracle residual threshold reaches 0.4606.

The central negative result is that the synthetic background+CNN pipeline does not beat a simple non-learning residual threshold under the strict causal protocol. Because the non-learning threshold is tuned on the test set, its advantage is an upper bound; the deep model's deficit is therefore conservative. Reliability gating, boundary attention and background-degradation simulation do not change this conclusion: the corresponding variants reach 0.240, 0.249 and 0.260, all

Table 2 Background-provenance switching at fixed weights. Four held-out categories, stride 1. The two rightmost columns are the gaps defined in Eq. (1).

Model	Leaky	Causal	First	Oracle	Δ_{leak}	Δ_{oracle}
Domain (`gnet_real`)	**0.6661**	**0.4526**	0.5132	0.6119	**+0.2135**	**+0.1594**
Synthetic (`gnet_main`)	0.3607	0.3297	0.3459	0.3506	+0.0310	+0.0209

Table 3 Strict-causal comparison on CDnet 2014 (53 videos, stride 1). The non-learning residual thresholds scan τ on the test set and are therefore optimistic. "rel" denotes a reliability-gated variant.

Method	Bg.	F1 (cat)	PWC (cat)
SuBSENSE (mask)	—	**0.7287**	1.7742
CFO network	none	0.2209	9.5100
Synth. GNet (3 seeds)	causal	0.2901 ± 0.0414	14.86
Synth. + rel. gate	causal/rel	0.2404	15.6061
Synth. + boundary	causal/rel	0.2491	17.0017
Synth. + full	causal/rel	0.2603	17.5604
Causal residual τ	causal	**0.3918**	—
First-frame res. τ	first	0.3973	—
Oracle residual τ	oracle	0.4606	—

below the synthetic baseline and far below the residual threshold. We are careful not to overgeneralize: the testbed is one architecture and one synthetic training recipe. But the result is sufficient to reject the strong claim that a synthetic background+CNN pipeline is already a better foreground segmenter than a causal residual baseline on CDnet. Any such claim needs a strict-causal comparison, not a comparison with a leaky or oracle pipeline. Per-class results (Appendix A) show that the remaining errors concentrate in PTZ, nightVideos, turbulence and dynamicBackground, where causal backgrounds are least reliable, while thermal and badWeather are comparatively strong. This is consistent with background quality, rather than architecture, driving the failures.

Domain and label diagnosis.

To test whether the bottleneck is the training domain or the label source, we trained a label-free variant on seven CDnet categories using causal-median teacher masks (no human annotation) and evaluated it on the same four held-out categories at stride 1. It reaches 0.417 at stride 1, up from 0.330 for the synthetic model and close to the 0.453 of the domain model trained with SuBSENSE pseudo-labels. At stride 5 the label-free student reaches 0.419, its causal-median teacher 0.465, and the domain model 0.453 on the same four categories. The label-free student therefore does not exceed its teacher: the paired difference is -0.044 (95% CI $[-0.134, +0.036]$), and it is statistically indistinguishable from the domain model at stride 5 (-0.030, $[-0.115, +0.056]$). All three remain far below SuBSENSE (0.726). For reference, the non-learning causal residual threshold reaches 0.373 (stride 1), so label-free training adds value over a simple residual, but not over its teacher, the domain model, or SuBSENSE. The result supports a domain/label diagnosis: real-domain data helps, whereas synthetic-to-real shift is a major bottleneck. It also shows that removing human labels does not by itself remove the gap to a strong BGS teacher.

4.5 RQ4: How large is seed variance?

The three seeds of the synthetic model give F1 = 0.3023, 0.3243 and 0.2438, with mean 0.290 and standard deviation 0.041. Many ablation differences reported in this literature are smaller than this. In our own reproduction, TSAM depth, data augmentation and background-model choices produce differences of 0.01–0.04, which fall inside the seed noise. A single run can therefore support a narrative that a multi-seed run does not. Repeated-run variance can change rankings [36], and reporting score distributions rather than single best values is recommended [37, 38]. We recommend reporting mean and standard deviation over at least three seeds, and treating differences smaller than the seed standard deviation as inconclusive.

4.6 RQ5: Cross-dataset transfer to SBI 2015

Table 4 reports the cross-dataset results. The synthetic model reaches 0.3468, while the non-learning causal residual threshold reaches 0.5888 and the first-frame residual reaches 0.6565. The domain model reaches 0.6647 and the oracle residual reaches 0.7091. The qualitative conclusion is

Table 4 Cross-dataset results on SBI 2015 (stride 1, causal background unless noted). No model is trained on SBI.

Method	Bg.	F1 (cat)
Synthetic GNet	causal	0.3468
Causal residual τ	causal	0.5888
First-frame res. τ	first	0.6565
Domain GNet	causal	0.6647
Oracle residual τ	oracle	0.7091

Table 5 Background-corruption sensitivity on four categories (PTZ, nightVideos, intermittentObjectMotion, thermal; stride 1). Ghost and illumination are relative perturbation strengths; noise is the standard deviation of additive Gaussian noise. F1 (category average).

Model	clean	ghost .30	illum .20	noise .10
Synthetic GNet	0.270	0.261	0.259	0.172
Domain (leaky)	**0.372**	**0.181**	0.350	**0.075**

the same as on CDnet: under a causal protocol, the synthetic background+CNN model is far below a simple residual baseline. The gap between the synthetic and domain models also shows that the synthetic-to-real domain shift, not the background estimator alone, is a major bottleneck. The domain model is trained on real CDnet data; its higher score is therefore not evidence that the synthetic recipe generalizes.

4.7 RQ6: Reliability gating, robustness and efficiency

The reliability-gated variant is a diagnostic for the claim that a learned per-pixel gate improves segmentation under background contamination. On the same four held-out categories, the learned-gate checkpoint reaches 0.280 and the full variant 0.313, compared with 0.270 for the ungated synthetic checkpoint and 0.214 for a training-free gate on the same synthetic weights. On all 53 videos, the learned gate reaches 0.240 and the full variant 0.260, both below the 0.302 of the synthetic model. The learned gate therefore does not show a consistent gain: its held-out value is within seed noise, while its full-set value is lower, and a training-free gate on fixed weights is clearly harmful. The gated variants use a reliability-gated background while the main model uses a causal running average, so the comparison is not a pure architecture ablation; nevertheless, no variant shows a robust positive effect. We therefore report the reliability-gating result as negative. This does not show that reliability estimation is useless; it shows that the specific learned gate does not improve accuracy consistently in our testbed.

To probe sensitivity to background quality, we perturb the causal background at test time with ghost residue, illumination drift and noise (Table 5). The domain model is the most sensitive: ghost 0.30 reduces F1 from 0.372 to 0.181 and noise 0.10 reduces it to 0.075, consistent with its reliance on the leaky background distribution seen in training. The synthetic model is more stable under ghost and illumination changes but starts from a lower level and also collapses under strong noise. These curves are diagnostic: they show that background quality changes the reading by a large margin, not that a particular module is robust.

For efficiency, the network has 60.6M parameters, 431.7 GMACs and runs at about 16 FPS at 513^2 with 0.82 GB peak memory. Background estimation costs 1.7 ms per frame, and the reliability map adds 4.2 ms per frame. These numbers describe the testbed, not a deployment claim.

Boundary F1 (2-pixel band, category average, strict-causal protocol) is 0.320 for the synthetic model, 0.300 with the reliability gate, 0.286 with boundary attention, 0.317 for the full variant and 0.360 for the current-frame-only network. The boundary-aware variants therefore do not improve BF1 in this testbed; the relatively high value of the current-frame-only network reflects its high boundary recall rather than better regions. We do not claim a boundary benefit.

5 Discussion and Limitations

5.1 What should be reported

Five reporting requirements follow. First, state the background provenance (causal/leaky/oracle); a dataset static background or whole-video median is an oracle. Second, report the teacher baseline on the same test set. Third, use a frozen protocol with fixed stride, region of interest, shadow handling and aggregation. Fourth, report seed variance and treat smaller differences as inconclusive. Fifth, report a non-learning residual baseline under the same protocol; if a deep model cannot beat it, that is a finding.

5.2 Interpretation

The domain and synthetic models fail in complementary ways. The domain model has a high leaky-background score but collapses under a causal background, and it does not beat its teacher. Its gain is therefore consistent with L1 and L2. The synthetic model is nearly invariant to provenance, which is desirable, but its absolute accuracy is low; the causal residual baseline beats it on both CDnet and SBI. The two results together show that the current paradigm conflates transferable foreground cues with estimator-specific information. Separating the two is a prerequisite for a credible generalization claim.

It is important to distinguish this critique from a blanket rejection of BGS. If the task is to post-process a BGS algorithm, a BGS-guided background is a legitimate input, and a high score is meaningful for that task. The problem is the mismatch between the design and the claim. A paper that claims to learn transferable foreground cues, or to generalize across scenes, must show that the gain survives when the BGS decision channel and future frames are removed. Our audit provides a simple way to run that check.

5.3 Limitations

This audit has several limitations. It is based on one GNet-style testbed and one synthetic training recipe; external validity requires testing more architectures and more background estimators. The controlled provenance-switching experiment uses four held-out categories and the same stride-1 protocol as the main table; all provenance levels use the same weights and frames, so the comparison is internally controlled. The non-learning baselines tune their threshold on the test set, which makes them optimistic; this is intentional, but it means their scores are upper bounds. The corruption table is diagnostic rather than a claim of robustness. The reliability-gating comparison uses a reliability-gated background rather than a pure architecture ablation. Finally, we evaluate on CDnet and SBI only; we do not claim universal generalization [43].

6 Conclusion

We presented an empirical audit of background-input provenance in background-guided video foreground segmentation. We formalized three leakage channels and the causal/leaky/oracle levels with two gap metrics. The domain model loses 0.214 F1 when the BGS-guided background is replaced by a causal one and does not beat its teacher; the oracle gap is 0.159 for the domain model and 0.021 for the synthetic model. Under a strict causal protocol, the synthetic background+CNN pipeline does not beat a non-learning causal residual threshold on CDnet or SBI, and seed variance exceeds most ablation differences. These results do not show that background-guided segmentation is impossible; they show that reported gains must be separated from leakage and self-distillation. We release the code, splits and per-video results so that the audit can be repeated.

A Per-class results

Table 6 Per-class F1 on CDnet 2014 (stride 1). bW: badWeather; base: baseline; cJ: cameraJitter; dB: dynamicBackground; IOM: intermittentObjectMotion; lF: lowFramerate; nV: nightVideos; sh: shadow; th: thermal; tur: turbulence.

Cat.	SuB	CFO	CRes	Syn	+Rel	Full
PTZ	0.3788	0.0362	0.2443	0.0877	0.0797	0.0658
bW	0.8576	0.3220	0.4270	0.3407	0.4278	0.4383
base	0.9491	0.1968	0.5891	0.3955	0.2109	0.1877
cJ	0.8090	0.2980	0.4034	0.3486	0.3123	0.3061
dB	0.8163	0.3072	0.2821	0.3131	0.2448	0.1396
IOM	0.6050	0.1830	0.2855	0.2165	0.1317	0.3968
lF	0.6567	0.0914	0.4635	0.1679	0.1421	0.1588
nV	0.4895	0.0342	0.3515	0.0930	0.0605	0.0666
sh	0.8994	0.3060	0.5491	0.4778	0.4030	0.3578
th	0.6860	0.4943	0.2680	0.6842	0.5924	0.6386
tur	0.8685	0.1605	0.4464	0.2010	0.0395	0.1071
Avg.	**0.7287**	**0.2209**	**0.3918**	**0.3023**	**0.2404**	**0.2603**

Table 6 reports the per-class F1 values for the strict-causal comparison. The causal residual column uses $\tau = 40$, the global optimum for the category average; the per-class optimum may differ. The synthetic, reliability and full columns are single-seed results for seed 0. The full column adds boundary attention and background-degradation simulation; all columns use the strict-causal protocol.

Supplementary information. The code, frozen protocol, video-level splits and per-video results are available in the accompanying repository.

Acknowledgements. Not applicable.

Statements and Declarations

- **Funding**: Not applicable.
- **Competing interests**: The author declares no competing interests.
- **Ethics approval**: Not applicable (no human or animal subjects).
- **Data availability**: The code, evaluation protocol, video splits and per-video results are available at https://github.com/TompsonL/gnet-leakage-audit. The datasets (CDnet 2014, SBI 2015, Cityscapes, COCO) follow their original licenses and are not redistributed.
- **Author contribution**: J. Liu conceived and conducted the study and wrote the manuscript.

References

[1] Wang, Y., Jodoin, P.-M., Porikli, F., Konrad, J., Benezeth, Y., Ishwar, P.: Cdnet 2014: An expanded change detection benchmark dataset. In: 2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops, pp. 393–400 (2014). https://doi.org/10.1109/cvprw.2014.126

[2] Stauffer, C., Grimson, W.E.L.: Adaptive background mixture models for real-time tracking. In: Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR), vol. 2, pp. 246–252 (1999). https://doi.org/10.1109/cvpr.1999.784637

[3] Barnich, O., Van Droogenbroeck, M.: Vibe: A universal background subtraction algorithm for video sequences. IEEE Transactions on Image Processing **20**(6), 1709–1724 (2011) https://doi.org/10.1109/tip.2010.2101613

[4] St-Charles, P.-L., Bilodeau, G.-A., Bergevin, R.: Subsense: A universal change detection method with local adaptive sensitivity. IEEE Transactions on Image Processing **24**(1), 359–373 (2015) https://doi.org/10.1109/tip.2014.2378053

[5] Hofmann, M., Tiefenbacher, P., Rigoll, G.: Background segmentation with feedback: The pixel-based adaptive segmenter. In: 2012 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops, pp. 38–43 (2012). https://doi.org/10.1109/cvprw.2012.6238925

[6] Kim, J.-Y., Ha, J.-E.: Foreground objects detection using a fully convolutional network with a background model image and multiple original images. IEEE Access **8**, 159864–159878 (2020) https://doi.org/10.1109/access.2020.3020818

[7] Braham, M., Van Droogenbroeck, M.: Deep background subtraction with scene-specific convolutional neural networks. In: 2016 International Conference on Systems, Signals and Image Processing (IWSSIP), pp. 1–4 (2016). https://doi.org/10.1109/iwssip.2016.7502717

[8] Babaee, M., Dinh, D.T., Rigoll, G.: A deep convolutional neural network for video sequence background subtraction. Pattern Recognition **76**, 635–649 (2018) https://doi.org/10.1016/j.patcog.2017.09.040

[9] Lim, L.A., Yalim Keles, H.: Foreground segmentation using convolutional neural networks for multiscale feature encoding. Pattern Recognition Letters **112**, 256–262 (2018) https://doi.org/10.1016/j.patrec.2018.08.002

[10] Lim, L.A., Keles, H.Y.: Learning multi-scale features for foreground segmentation. Pattern Analysis and Applications **23**(3), 1369–1380 (2020) https://doi.org/10.1007/s10044-019-00845-9

[11] Tezcan, M.O., Ishwar, P., Konrad, J.: Bsuv-net: A fully-convolutional neural network for background subtraction of unseen videos. In: 2020 IEEE Winter Conference on Applications of Computer Vision (WACV), pp. 2763–2772 (2020). https://doi.org/10.1109/wacv45572.2020.9093464

[12] Tezcan, M.O., Ishwar, P., Konrad, J.: Bsuvnet 2.0: Spatio-temporal data augmentations for video-agnostic supervised background subtraction. IEEE Access **9**, 53849–53860 (2021) https://doi.org/10.1109/access.2021.3071163

[13] Chung, N.M., Ha, S.V.-U.: Bgsubnet: Robust semisupervised background subtraction in realistic scenes. IEEE Sensors Journal **24**(6), 9172–9186 (2024) https://doi.org/10.1109/jsen.2024.3358181

[14] An, Y., Zhao, X., Yu, T., Gu, H., Zhao, C., Tang, M., Wang, J.: Zbs: Zero-shot background subtraction via instance-level background modeling and foreground selection. In: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 6355–6364 (2023). https://doi.org/10.1109/cvpr52729.2023.00615

[15] Cho, S., Lee, M., Lee, J., Cho, M., Park, S., Kim, J., Jang, H., Lee, S.: Treating motion as option with output selection for unsupervised video object segmentation. IEEE Transactions on Circuits and Systems for Video Technology **35**(10), 10278–10292 (2025) https://doi.org/10.1109/tcsvt.2025.3562186

[16] Alawode, B., Javed, S.: Learning spatial–temporal regularized tensor sparse rpca for background subtraction. IEEE Transactions on Neural Networks and Learning Systems **36**(6), 11034–11048 (2025) https://doi.org/10.1109/tnnls.2025.3543947

[17] Alawode, B., Javed, S.: Unsupervised background subtraction using generator-discriminator learning. IEEE Transactions on Circuits and Systems for Video Technology **36**(1), 986–1002 (2026) https://doi.org/10.1109/tcsvt.2025.3598078

[18] Sauvalle, B., La Fortelle, A.: Autoencoder-based background reconstruction and foreground segmentation with background noise estimation. In: 2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 3243–3254 (2023). https://doi.org/10.1109/wacv56688.2023.00326

[19] Elgammal, A., Harwood, D., Davis, L.: Non-parametric model for background subtraction. In: Computer Vision – ECCV 2000, pp. 751–767 (2000). https://doi.org/10.1007/3-540-45053-x_48

[20] Goyal, K., Singhai, J.: Review of background subtraction methods using gaussian mixture model for video surveillance systems. Artificial Intelligence Review **50**(2), 241–259 (2017) https://doi.org/10.1007/s10462-017-9542-x

[21] Maddalena, L., Petrosino, A.: A self-organizing approach to background subtraction for visual surveillance applications. IEEE Transactions on Image Processing **17**(7), 1168–1177 (2008) https://doi.org/10.1109/tip.2008.924285

[22] Bouwmans, T.: Traditional and recent approaches in background modeling for foreground detection: An overview. Computer Science Review **11-12**, 31–66 (2014) https://doi.org/10.1016/j.cosrev.2014.04.001

[23] Garcia-Garcia, B., Bouwmans, T., Rosales Silva, A.J.: Background subtraction in real applications: Challenges, current models and future directions. Computer Science Review **35**, 100204 (2020) https://doi.org/10.1016/j.cosrev.2019.100204

[24] Sobral, A., Vacavant, A.: A comprehensive review of background subtraction algorithms evaluated with synthetic and real videos. Computer Vision and Image Understanding **122**, 4–21 (2014) https://doi.org/10.1016/j.cviu.2013.12.005

[25] Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., Wichmann, F.A.: Shortcut learning in deep neural networks. Nature Machine Intelligence **2**(11), 665–673 (2020) https://doi.org/10.1038/s42256-020-00257-z

[26] Lapuschkin, S., Wäldchen, S., Binder, A., Montavon, G., Samek, W., Müller, K.-R.: Unmasking clever hans predictors and assessing what machines really learn. Nature Communications **10**(1), 1096 (2019) https://doi.

org/10.1038/s41467-019-08987-4

[27] Kaufman, S., Rosset, S., Perlich, C., Stitelman, O.: Leakage in data mining. ACM Transactions on Knowledge Discovery from Data **6**(4), 1–21 (2012) https://doi.org/10.1145/2382577.2382579

[28] Kapoor, S., Narayanan, A.: Leakage and the reproducibility crisis in machine-learning-based science. Patterns **4**(9), 100804 (2023) https://doi.org/10.1016/j.patter.2023.100804

[29] Xie, Q., Luong, M.-T., Hovy, E., Le, Q.V.: Self-training with noisy student improves imagenet classification. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 10684–10695 (2020). https://doi.org/10.1109/cvpr42600.2020.01070

[30] Arazo, E., Ortego, D., Albert, P., O'Connor, N.E., McGuinness, K.: Pseudo-labeling and confirmation bias in deep semi-supervised learning. In: 2020 International Joint Conference on Neural Networks (IJCNN), pp. 1–8 (2020). https://doi.org/10.1109/ijcnn48605.2020.9207304

[31] Gawlikowski, J., Tassi, C.R.N., Ali, M., Lee, J., Humt, M., Feng, J., Kruspe, A., Triebel, R., *et al.*: A survey of uncertainty in deep neural networks. Artificial Intelligence Review **56**(S1), 1513–1589 (2023) https://doi.org/10.1007/s10462-023-10562-9

[32] Goyette, N., Jodoin, P.-M., Porikli, F., Konrad, J., Ishwar, P.: Changedetection.net: A new change detection benchmark dataset. In: 2012 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops, pp. 1–8 (2012). https://doi.org/10.1109/cvprw.2012.6238919

[33] Maddalena, L., Petrosino, A.: Towards benchmarking scene background initialization. In: Image Analysis and Processing – ICIAP 2015 Workshops, pp. 469–476 (2015). https://doi.org/10.1007/978-3-319-23222-5_57

[34] Bouwmans, T., Maddalena, L., Petrosino, A.: Scene background initialization: A taxonomy. Pattern Recognition Letters **96**, 3–11 (2017) https://doi.org/10.1016/j.patrec.2016.12.024

[35] Csurka, G., Larlus, D., Perronnin, F.: What is a good evaluation measure for semantic segmentation? In: Proceedings of the British Machine Vision Conference (BMVC), pp. 32–13211 (2013). https://doi.org/10.5244/c.27.32

[36] Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., Meger, D.: Deep reinforcement learning that matters. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 32 (2018). https://doi.org/10.1609/aaai.v32i1.11694

[37] Reimers, N., Gurevych, I.: Reporting score distributions makes a difference: Performance study of LSTM-networks for sequence tagging. In: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing, pp. 338–348 (2017). https://doi.org/10.18653/v1/d17-1035

[38] Dodge, J., Gururangan, S., Card, D., Schwartz, R., Smith, N.A.: Show your work: Improved reporting of experimental results. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 2185–2194 (2019). https://doi.org/10.18653/v1/d19-1224

[39] Caelles, S., Maninis, K.-K., Pont-Tuset, J., Leal-Taixe, L., Cremers, D., Van Gool, L.: One-shot video object segmentation. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 5320–5329 (2017). https://doi.org/10.1109/cvpr.2017.565

[40] Perazzi, F., Pont-Tuset, J., McWilliams, B., Van Gool, L., Gross, M., Sorkine-Hornung, A.: A benchmark dataset and evaluation methodology for video object segmentation. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR),

pp. 724–732 (2016). https://doi.org/10.1109/cvpr.2016.85

[41] Cordts, M., Omran, M., Ramos, S., Rehfeld, T., Enzweiler, M., Benenson, R., Franke, U., Roth, S., *et al.*: The cityscapes dataset for semantic urban scene understanding. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3213–3223 (2016). https://doi.org/10.1109/cvpr.2016.350

[42] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft coco: Common objects in context. In: Computer Vision – ECCV 2014, pp. 740–755 (2014). https://doi.org/10.1007/978-3-319-10602-1_48

[43] Zhou, K., Liu, Z., Qiao, Y., Xiang, T., Loy, C.C.: Domain generalization: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **45**(4), 4396–4415 (2023) https://doi.org/10.1109/tpami.2022.3195549